



Seaborn Linear Regression Models Transcript

Hello everyone, we'll now understand how to fit a regression model in the Python environment using the Seaborn libraries. Okay, what is regression? Regression is a statistical technique wherein you understand the relationship between dependent and independent variables. In general, when we understand certain phenomenon, let us say you want to understand weight of a person, you think his height might be a determining variable.

Underlying hypothesis being the person with a greater height is expected to have a greater weight. Similarly, family which is earning more or having a larger income is expected to have a higher consumption. These are relations that are known to everyone and the standard correlation plot would explain that relationship between the two variables.

Now, the question is what is the extent of influence that a dependent variable has on the independent variable is given by a regression model. However, the understanding here is to capture the coefficient of regression so that you can predict the future values using that coefficient. Now, for the purposes of this course, we are not getting into the details, statistical details of the modelling details of the regression models, but we are focused on how to present the visualisations regarding to regression because in certain contexts, especially in the financial space, if you are developing models as part of your larger course or in this particular BFSS specialisation, you might need regression models.

You will definitely use regression models in many contexts and you might want to convey the results to your top management. In that case, you want to use the visualisations on presenting the regression results and actual regression lines so that there is more acceptance in the audience. So, we will now try to use the Seaborn library quickly to understand the features of regression and then see how to plot a regression diagram.

So, for that we are importing the libraries that are required. We already have Seaborn and Matplotlib, but we are now installing stat models so that you can run a regression. Of course, we are also importing NumPy and Pandas and I have uploaded the TIPS dataset which you have already seen and the regression plot which is given by the command `sns.reg` plot tries to portray the relationship between X total bill and Y tip and the data is TIPS.

So, the relationship between the total bill amount and the tip received is plotted and a regression line is superimposed on that. Now, let us understand what does this mean and this is the plot that is produced. Now, I will take you through some steps so you will understand the importance of a regression plot.

So, I am just doing a simple cat plot between total bill amount and tip which should give me a scatter diagram telling the relationship between the total bill and the tip amount. So, you should get a scatter plot which should look like this same diagram without the line. So, this is the diagram.

Okay. So, you got the diagram. You have the total bill amount and it is showing the total bill amount and total tip amount.

This is how the distribution is. Okay. `sns.cat` plot.

This is the relationship between total bill amount. So, as the bill amount increases, the tip amount is increasing. So, what is this particular regression plot or regression line is doing is it is trying to fit a relationship between total bill amount and the tip.

So, this straight line represents that equation where if I give you the total bill amount you should be able to tell the corresponding tip amount. Okay. It need not be the actual data but it represents a line which gives a overall representation of the existing underlying relationship between total bill and tip amount.

Okay. So, regression line does not pass through all the points but it captures the essence of all these points. So, interested people can learn about regression on their own but for the purposes of this particular course we are interested in plotting the regression line and the regression relationship between the dependent variable which is the tip amount and the independent variable which is the total bill.

Larger is the total bill you expect the larger the tip amount to be. So, total bill is influencing or determining the tip amount. Therefore, total bill is the independent variable and tip amount is the dependent variable.

Okay. So, you can use the `red plot` command for plotting total bill versus tip and you will get a nice graph like this and the thick blue line indicates the regression line which captures the linear relationship between total bill and tip amount. Now, let us look at the another function called `SMS SNS LM` plot.

Okay. So, it also produces a similar result. It is a different command.

Sorry. It produces the same result giving the regression line capturing the relationship between total bill and tip. So, what is the difference between `LM` plot and the regression line? A `red plot` function.

`Red plot` is an access level function while `LM plot` is a figure level function. Therefore, `LM plot` is more flexible format of the data. Right.

LM plot is more flexible. So, sorry. Red plot is more flexible for about the data format it accepts whereas LM plot requires X and Y to be strings referring to columns in the particular data frame.

So, these are functional differences but one more advantage here is with LM plot has the hue parameter. So, what I can do I can take LM plot and parse certain commands to overlay additional data points like I can set the hue in this case to be the gender which we have seen earlier. Right.

So, if I do this I am getting two regression plots. Right. For one for the male one for the female both present in the same diagram.

So, instead of that I can try to do them in the different columns so that they look more pleasing and appropriate. So, this is the regression plot for male customers. This is the regression plot for the female customers and now maybe either want to do that also with the time of the lunch.

Right. When the time of the visit to the restaurant. So, now you can see the difference in which you can use the parameters and bring out more relationship.

So, now let us understand what is this four diagrams are meaning. For male customers and during the dinner time this is the regression function and for the female female customers during lunch time this is the regression function. So, you can estimate four different relationships and show their impact in a single graph by imposing more dimensions and enhance in the richness of your graphic.

Now, let us talk about different kinds of models. We just plotted a linear function. We can also plot a nonlinear function so that you can have a better fit of the data.

So, let me explain that. If you remember the Anscombe's quadrat which shows that the central tendencies are the aggregate measures of the data are similar for four different data sets but the underlying data distributions are very different. Right.

You had a highly distributed data. You had a parabolic data. You had you had a parabolic data and you had a single out layer.

There are different kinds of data which had the same central tendency measures like the mean median. Right. So, this is what the speciality about Anscombe format, Anscombe's quadrat and we have used that as a case showing that you need to visualise data to understand how is it what is the pattern.

So, now let us load Anscombe data and see what regression functions give better results for each data set. So, first we are trying to plot the relationship between the first data set. Okay.



Let me first before I do anything let us look at Anscombe. Okay. Dot head.

So, this is XY of each data 1 2 3 4 the day. Okay. I should also do the shape to understand the total shape.

Yeah. So, there are 44 rows and 3 columns. Okay.

So, let us look at all the 44. So, yeah. How is the data? It is categorised as whether the data set is belonging to first, second, third and fourth.

You have four data sets. Right. And about 10 observations in each or 11 observations in each.

Okay. That is how you have the data and on X and Y variables. Right.

This is Anscombe quadrat and just to visualise what we can do the Anscombe quadrat. Right. I am just doing SMS LM plot data is equal to Anscombe and I am subsetting the data to be the first category.

Okay. Within the data set to be first. Then I am setting the confidence interval to be none which is nothing but the yellow the blue band around the linear regression line showing the confidence percentage around the line.

Right. And the scatter. So, this is what we have because we wanted to highlight the scatter a size to be 80.

Okay. If you can reduce or increase the size I have to increase the size just to show the difference. Okay.

Maybe I can make it even more 20 also. So, you get the smaller dots. So, let us keep a larger dot.

So, the visibility is clear. So, this is how you can you have plotted for one data set. Okay.

So, this is the linear plot now. Okay. It is fitting reasonably well.

Now, let us try this for plotting the data set 2 which looks like a parabola. Right. Which looks like a parabola.

So, I am just using this and I am plotting the data. So, you can clearly see the straight line or linear function LM plot is not most ideal for data set 2 which has a parabolic



shape or which has a curvature and the linear straight line is failing to capture it completely. So, what we try to do is we try to fit a polynomial function.

Okay. Of order 2. Okay. We try to incorporate.

If you look at the difference between this and this, we are just trying to fit a polynomial function of order 2. Okay. Let me show what happens if you just plot a polynomial of order 1. You will get the same result which is the straight line. Now, instead of plotting a polynomial of order 1, I am plotting a polynomial of order 2. Right.

So, you should be getting a curvature and that should be able to capture the data ideally. So, now you have a plot which captures the data very well because it is using a polynomial function instead of a linear function and you can experiment with these things. Now, let us look at the data set 3. Okay.

Data set 3 and then it uses. So, now let us try to fit the plot, linear plot for all the elements in the Anscombe query, Anscombe squared rate. I just use the column.

I set the column to be data set. So, it will have four different graphs because there are four different data sets just running this. So, if you fit, forcefully fit the linear plot on all Anscombe squared rate, you might not have the best ideal fit in each of the cases.

Therefore, it is better to modify the functions for different cases. Okay. And we have also seen how to use the hue characteristic and impose the additional dimensions on the regression plot.

Okay. Thank you.

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